PRODUCTION READY

October 2026 • Executive Status Report

Platform Modernization & Engineering Report

A concise record of all upgrades, new features, and infrastructure improvements delivered on the Edukom Digital Ecosystem.

Strategic Outcome: Edukom Group has transitioned from a fragmented collection of manual spreadsheets and external subscriptions (TutorCruncher, Zoho, Zoom, Google Meet) into a **fully unified, proprietary digital education ecosystem**. The platform now delivers automated tutor payroll, embedded virtual classrooms, encrypted video courses, multi-currency tuition billing, automated WhatsApp & SMS alerts, and dedicated portals for students, parents, tutors, and management.

100%	0 CSVs	5 Portals	99.9%
TEST PASS RATE (72/72)	DIRECT LMS-TO-PAYROLL	STUDENT, TUTOR, PARENT, STAFF, ADMIN	TARGET SERVICE UPTIME

1. Platform Evolution (Before vs. Now)

Area	Legacy Setup (At Project Start)	Modern Edukom Platform (Delivered)
Live Classrooms	× Sent loose Zoom/Google Meet links over WhatsApp; no attendance logs.	✓ **Embedded WebRTC Classrooms** with collaborative whiteboard, screen share, and instant join.
Tutor Payroll	× Manual spreadsheets; tedious calculations; high risk of spreadsheet calculation errors.	✓ **Automated 140-Tier Payroll Engine** generated directly from completed classes with 1 click.
Bank Account Verification	× Manual review of account numbers; frequent bank transfer delays and errors.	✓ **Real-Time NUBAN Bank Validation** confirming tutor account names before disbursement.
Tuition Payments	× Manual bank transfers and screenshot verification.	✓ **Multi-Currency Online Gateway** (Paystack & Flutterwave) accepting Naira, USD, GBP, and EUR.
Recorded Lessons	× Raw unencrypted video files shared on Google Drive; content easily leaked.	✓ **Anti-Piracy Video Library (DRM)** with viewer watermarks and single-device login limits.
Reminders & Alerts	× Operations manually typed WhatsApp messages to parents.	✓ **Automated 24h, 1h, and 10m WhatsApp/SMS alerts** sent to students, tutors, and parents.
Parent Experience	× Parents had no dashboard; had to inquire constantly for updates.	✓ **Dedicated Guardian Portal** with a multi-child switcher, attendance stats, and 1-click invoice payments.

Area	Legacy Setup (At Project Start)	Modern Edukom Platform (Delivered)
Student Assessments	× Ad-hoc homework over email; no verifiable certificates.	✓ **Interactive Self-Grading Quizzes**, homework submission drawers, and **QR-verifiable certificates**.
Mobile Experience	× Desktop-centric layout; difficult to navigate on mobile phones.	✓ **Mobile-First Native App Feel** with bottom navigation dock and quick-start onboarding checklist.

2. Key Upgrades Delivered by Operational Area

A. Virtual Classrooms & Scheduling Completed

- **Embedded Video Rooms:** High-definition video with in-room chat, screen sharing, hand-raising, and live digital whiteboard.
- **No Link Sharing:** Rooms are private and token-guarded; students enter directly from their dashboard.
- **Double-Booking Guard:** Prevents tutor and student timetable clashes automatically.
- **Dual-Timezone Clock:** Displays both local time and West Africa Time (WAT) for diaspora students in the UK, US, and UAE.
- **Calendar Sync:** 1-click export to Google, Apple, and Outlook calendars via `.ics` feed.

B. Automated Payroll & Bank KYC Completed

- **Zero-CSV Direct Bridge:** Operations can generate an entire monthly payroll run directly from completed lessons with 1 click.
- **140 Compensation Tiers:** Automatically calculates exact earnings across subjects, curricula, and experience levels.
- **Instant NUBAN Bank Resolution:** Validates Nigerian bank accounts in real time against official banking databases.
- **Self-Service Payslips:** Tutors can view, verify, and download itemized compensation statements on their dashboard.
- **Cybersecurity Shield:** Formulas in spreadsheets are automatically neutralized against security vulnerabilities (CWE-1236).

C. Anti-Piracy Video Library & Studio Completed

- **Encrypted Video Player (DRM):** High-grade AES-128 encryption prevents unauthorized ripping and screen downloading.
- **Concurrent Stream Guard:** Restricts accounts to 1 device at a time, eliminating illegal credential sharing.
- **Dynamic Watermark:** Displays the viewer's registered name and IP address on screen during playback to deter leaks.
- **Adaptive Data Saver Mode:** Automatically adjusts video quality (down to 360p) for smooth playback on mobile networks.
- **Creator Studio:** Instructors can create video courses, structure modules, and upload worksheets easily.

D. WhatsApp & SMS Automations Completed

- **3-Tier Multi-Stakeholder Reminders:** Automated reminders sent at **24 hours**, **1 hour**, and **10 minutes** before class to students, tutors, and parents.
- **Subscription Expiry Alerts:** Proactive renewal reminders sent 3 days and 1 day prior to package expiration.
- **Two-Way Parent Triage:** Parents can reply to WhatsApp reminders; replies are logged and urgent keywords (e.g. *cancel*, *emergency*) trigger immediate alerts to staff.
- **Admin Operations Hub:** Real-time monitoring dashboard with queue inspection and test simulation modes.
- **Parent Quiet Hours:** Respects family privacy by suppressing non-urgent notifications during night hours.

E. Dedicated Portals & Academic Tools

Completed

- **Guardian Portal:** Multi-child switcher allowing parents to track attendance, grades, and tutor notes for each child.
- **Student Homework Drawer:** Students submit homework assignments (essays & files); tutors grade directly in their portal.
- **Interactive Quizzes:** Instant self-grading multiple choice quizzes with detailed explanations.
- **Verifiable Digital Certificates:** Branded graduation certificates with unique QR codes for public verification.
- **Tutor Hub:** Daily teaching schedule, pending grading alerts, and accrued earnings metrics.

F. Mobile-First UX & Guided Onboarding

Completed

- **Mobile Bottom Navigation Dock:** Native app-like quick access to Home, Courses, Get Tutor, and Videos.
- **Guided 4-Step Onboarding:** Interactive progress bar guiding new students and tutors through account setup.
- **Classroom Hardware Diagnostic:** Built-in microphone and camera tester ensuring zero technical hitches during class.
- **Returning User Hub:** Welcome-back streak tracking and a 1-click countdown button directly into today's next lesson.
- **Branded Sign-In Redesign:** Clean, responsive sign-in and sign-up interfaces across LMS and HR portals.

3. System Health & Production Readiness

Quality Assurance & Testing: The entire platform was validated through a comprehensive automated verification suite: **All 72 automated unit tests passed with a 100% success rate** spanning payments, payroll calculations, classroom security, DRM encryption, and automations. The underlying production infrastructure is fully containerized (Docker & Docker Compose), fortified with secure Nginx reverse proxy configurations, daily encrypted database backups, and health monitoring endpoints (`/healthz`) committed to a **99.9% uptime SLA**.